

Problem Set 7

POLS W4732

Fall 2026

Due December 4 before the TA section

Artificial-intelligence policy. Unless an exercise explicitly states otherwise, students may not use generative-AI systems (including ChatGPT, Claude, Gemini, Copilot, or AI-generated search answers) to formulate, derive, solve, explain, or write answers to this problem set. Course materials, ordinary calculators, typesetting software, the teaching team, and collaboration within the registered problem-set group are permitted. Any approved computational tool must be disclosed. Every group member is responsible for understanding and being able to explain the entire submission.

Exercise 1 (Policy competition). *There are three equally sized groups, R, M, P . A member of group J obtains*

$$U_J(p, s) = p(\alpha - p) + \beta_J s,$$

where $\beta_R = 0$, $\beta_P = 1$, $\beta_M \in (0, 1)$, $\alpha \in (1, 2)$, and $p + s \leq 1$. Candidates A and B commit to platforms; A has group R 's preferences and B has group P 's.

Part A: deterministic benchmark. *For Part A only, candidates maximize their probability of winning.*

- 1. Derive each group's ideal policy and verify directly that the budget binds and induced preferences over p are single-peaked.*
- 2. A proposed proof invokes the median-voter theorem without checking those conditions. Repair the proof that office-motivated candidates converge to group M 's ideal policy and state all needed assumptions.*

Part B: policy motivation and uncertain turnout. *Draw $X \sim U[0, 2]$. Exactly $X/2$ members of group M and $X/2$ members of group P vote, while all members of group R vote. Starting from convergent platforms, calculate candidate A 's payoff from a deviation to A 's ideal policy and from one intermediate policy. Characterize the full set of profitable deviations rather than merely exhibiting one. Repeat for the assigned value $\beta_M \in \{1/4, 1/2, 3/4\}$.*

Part C: probabilistic voting. *All voters turn out. A common shock $\xi \sim U[-1/(2\psi), 1/(2\psi)]$ favors candidate B ; group J votes for B iff $V_J(p_A) \leq V_J(p_B) + \xi$.*

- 1. State restrictions ensuring interior voting probabilities and derive candidate A 's winning probability $\pi(p_A, p_B)$.*
- 2. Derive both candidates' objectives and first-order conditions.*

3. *Prove the location and slope properties of both best responses, checking second-order or global-optimality conditions.*
4. *For $\alpha = 2$ and $\beta_M = 1/2$, derive equilibrium platform divergence as a function of ψ .*
5. *Determine which conclusions survive if the common shock is logistic rather than uniform.*

Your submission must identify one step at which a local first-order condition alone is insufficient and supply the missing argument.

Exercise 2 (Cheap talk). *In the baseline Crawford–Sobel model, the state ω is uniform on $[0, 1]$, the Receiver’s ideal action is ω , and the Sender’s ideal action is $\omega + b$.*

A proposed proof says: “Because the cutoff $\omega^ < 1$, the Sender’s payoff is higher under the two-report equilibrium for every state.” Determine whether this pointwise claim is correct and locate any states where it fails. Compute the Sender’s ex ante expected payoff under babbling and under a two-bin equilibrium, using the equilibrium cutoff condition. Characterize the values of b for which the two-bin equilibrium exists. State separately what can be concluded pointwise, interim by message, and ex ante.*